Circular motion

antocciphysics | TEACHER KEY | CPA

Learning target: Relate inward net force to a change in velocity direction.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 9; 3 course days later.*
Equal fall times; faster ball travels 5 times farther horizontally.

02. Explain a worked example (worked)
Velocity tangent at each point; net force toward center, perpendicular to instantaneous velocity.

03. Look and explain (model)
Velocity tangent; acceleration inward. Gravity, tension, friction, etc depending context.

04. Build the solution (practice)
6 N; doubles to 12 N.

05. Use the idea (reason)
Initially along tangent, not radially outward; subsequent forces may change path.

06. Choose a method (mixed) *Review: Unit 1, Day 20; 14 course days later.*
$a=10$ m/s² right; $v_f=70$ m/s right. Constant net force and mass throughout.

07. Exit check (exit)
No; direction changes.

08. STRETCH: Apply it another way (stretch)
Force becomes four times as large because speed is squared; 6 N becomes 24 N.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 9; 3 course days later.*
Speed 15 m/s. Need an initial position plus time and motion history/conditions; velocity alone is insufficient.

02. Explain a worked example (worked)
Velocity tangent at each point; net force toward center, perpendicular to instantaneous velocity.

03. Represent and interpret (model)
Velocity tangent; acceleration inward. Gravity, tension, friction, etc depending context.

04. Solve and justify (practice)
a 8 m/s², F 4 N. Doubling v quadruples: a 32, F 16 N.

05. Analyze the reasoning (reason)
Initially along tangent, not radially outward; subsequent forces may change path.

06. Choose a method (mixed) *Review: Unit 1, Day 20; 14 course days later.*
x interval [-88, -72] km; y interval [90, 110] km. Simplified geometry, gravity, Earth motion or missing observations limit the model. Bounds are not probabilities.

07. Exit check (exit)
Both halve; inverse proportional to radius.

08. STRETCH: Apply it another way (stretch)
$v_{max}=\sqrt{(Fr/m)}=4$ m/s; at 2 m, $\sqrt{32}\approx5.66$ m/s.